Wavefront Pruning in Budgeted Brownian Races

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Abstract

In a budgeted Brownian race, a controller watches a cloud of Brownian paths, pays per unit time for every path kept alive, may irreversibly prune, and spends a finite path-time budget to maximize the expected terminal maximum. We solve the mean-field version: a Poissonized population under per-path controls, with the budget enforced in expectation. The expected maximum has an exact layer-cake form whose gradient is a pivotal value, and for a fixed price on path-time the linearized best response is an obstacle problem whose keep sets are upper tails: the *wavefront policy*, which retains a path while its propagated future pivotal value exceeds its carrying cost. A fully corrective Frank–Wolfe method over wavefront policies, with dual bisection on the price, solves the budget problem and certifies its answer, bounding suboptimality by the Frank–Wolfe gap plus a complementary slackness term. In the committed example the adaptive policy improves the expected maximum by 19% over the best single screening date and by 30% over static thinning at the same expected budget; the finite-n race with a pathwise budget is left to future work.

1 Introduction

A search that runs many exploratory processes in parallel faces one recurring decision: which of them to keep paying for. The processes drift, their ranking changes, killing is irreversible, and only the best survivor is paid at the end. This paper studies the cleanest continuous model of that decision, which we call a *budgeted Brownian race*: a controller observes Brownian paths, pays per unit time for each surviving path, may irreversibly prune, and allocates a finite path-time budget to maximize the expected terminal maximum over a deterministic fallback.

The distinctive object is the *wavefront*: the narrow region near the future head of the race where one more path can still be terminally pivotal. The optimal policy has a one-line description: retain a path while its propagated future pivotal value exceeds its carrying cost. The body of the paper makes the description exact in the mean-field limit and computable at scale.

The name is chosen against three neighbors. Branching Brownian motion with selection retains a prescribed number of rightmost particles; here nothing branches, the survivor count is endogenous, and the resource is cumulative path-time rather than a headcount. Finite-fuel stochastic control is the classification home of the constraint, not a description of the terminal-maximum payoff. Beam search fixes its width in advance; wavefront pruning lets the width float under the budget.

2 The model

Optional paths form a Poisson cloud with initial intensity measure μ_0, and each alive path follows an independent Brownian motion. At each decision time a path is either killed irreversibly or kept for one more interval; keeping one path for dt costs dt. The budget B constrains expected aggregate path-time. The terminal payoff is the maximum of a deterministic fallback f and the surviving paths.

Controls are per-path and population-blind: each path's keep-or-kill decision depends on its own state and time, plus independent randomization, never on the realized population. Under this restriction the surviving cloud remains Poisson at every date, by independent thinning, and the terminal intensity m is the object of choice.

The restriction has a price a one-step example makes exact. Start Poisson(100) paths at zero with budget one and fallback zero. Independent thinning to intensity one pays about 0.3469; a controller who sees the population and keeps exactly one path whenever any exist pays $(1 - e^{-100})/\sqrt{2\pi} \approx 0.3989$ on the same expected budget. The values in this paper are optimal within the population-blind class.

3 The value of the terminal cloud

Fix a terminal grid $x_1 < \cdots < x_J$ and terminal intensity m with upper-tail intensity $R_j = \sum_{k \geq j} m_k$. The expected maximum is exactly

$$J(m) \;=\; f + \sum_{x_j > f} \Delta_j \left(1 - e^{-R_j}\right), \tag{1}$$

where Δ_j is the gap to the preceding attainable level. In English: each layer above the fallback pays its width times the probability that at least one survivor clears it. J is concave in m, and its gradient is the *pivotal value*

$$\frac{\partial J}{\partial m_j} \;=\; \mathbb{E}\big[(x_j - M)^+\big], \tag{2}$$

the expected improvement from placing one more infinitesimal path at x_j, where M is the current maximum.

4 Wavefront policies

For a shadow price $\lambda \geq 0$ on path-time, linearizing J at the current intensity gives a one-particle obstacle problem,

$$v_N = \mathbb{E}[(x - M)^+], \qquad v_k(x) = \max\big(0,\; \mathbb{E}[\, v_{k+1}(X_{k+1}) \mid X_k = x\,] - \lambda\, dt\big), \tag{3}$$

whose keep set at each step is an upper tail of the state space: one kill-below cutoff per time step. We call this the wavefront policy. The optimal population policy is in general a mixture of a few wavefront atoms, and the mixture is implementable per path: each path independently follows one atom, and by Poisson thinning the population intensity is exactly the mixture's.

5 The algorithm and its certificate

A fully corrective Frank–Wolfe (column-generation) method repeatedly computes the pivotal-value gradient (2), solves the obstacle problem (3) for the best-response wavefront policy, and re-optimizes the mixture of active atoms; bisection in λ enforces the budget. The scheme is a discrete counterpart of the coupled obstacle/Fokker–Planck KKT system.

The Frank–Wolfe gap alone does not certify accuracy at the requested budget, because the returned policy may underspend. The correct bound carries a complementary slackness term.

Lemma 1 (Budget suboptimality certificate). Let $C(m)$ denote expected path-time, $V(B) = \max\{J(m) : C(m) \le B\}$, and let m be any feasible intensity, with g the Frank–Wolfe gap of the Lagrangian $J - \lambda C$ at m. Then for every $\lambda \ge 0$,

$$V(B) - J(m) \;\le\; g + \lambda\,(B - C(m)).$$

Proof. By concavity of J, the Lagrangian optimum satisfies $L^\star \le L(m) + g$. For any feasible m', $J(m') \le L(m') + \lambda B \le L^\star + \lambda B \le L(m) + g + \lambda B$, and $L(m) = J(m) - \lambda C(m)$. □

The implementation returns a policy with $C(m) \le B$, thinning with the kill-all policy when the bisection overspends, and reports the two terms of the certificate separately. The slackness term is not small in practice: in one representative solve it exceeds the gap by a factor of seventy.

6 Related work

Selection in branching systems. Brunet and Derrida (1997) quantified what truncating a population does to its leading edge; Berestycki et al. (2013) worked out absorption at a moving barrier as a selection mechanism; Maillard (2016) pinned the speed and fluctuations of N-particle branching Brownian motion under spatial selection. In all three the population size is exogenous. The budgeted race replaces the fixed-N constraint with cumulative path-time, and nothing branches.

Budgets as fuel. Beneš et al. (1980) solved the original finite-fuel control problems in closed form. The label classifies the constraint here, a bounded cumulative control effort, but not the payoff.

Search, indices, and optionality. Weitzman (1979) opened boxes in reservation-value order; Gittins (1979) priced the next unit of attention with an index; Audibert et al. (2010) studied pure exploration under a hard budget of pulls. The bandit literature closest in payoff is the max-k-armed bandit of Cicirello and Smith (2005) and the extreme bandits of Carpentier and Valko (2014), where the criterion is the maximum reward found rather than the sum. The race is their continuous cousin: arms drift as diffusions, attention is metered in path-time, and killing is irreversible.

Mean-field control. Lasry and Lions (2007) coupled a value function to a density through a consistency condition; Carmona and Delarue (2018) treat control of McKean–Vlasov dynamics, the honest label for the population problem here. The race's version replaces the HJB by an obstacle problem, since the only control is stopping.

The algorithmic tools. Frank and Wolfe (1956) linearize and call an oracle; Jaggi (2013) supplies the duality-gap toolkit. The oracle here is the obstacle problem, and the atoms it returns are wavefront policies.

Budgeted rollouts of language models. Best-of-n sampling is a budgeted race: partial generations drift, compute is metered per token kept alive, and only the best completion is paid (Chen et al., 2021). Sun et al. (2024) kill low-reward generations mid-decode and reach best-of-n quality at a fraction of the cost; Bi et al. (2026) learn a prefix-level pruning signal for parallel reasoning under a fixed budget; Kalayci et al. (2025) stop sequential generation by Pandora's-box reasoning; Nomand et al. (2026) allocate a rollout budget by a two-threshold stopping rule. Each learns, scores, or proves its kill rule in a one-shot or sequential relaxation. None models the drifting value of a partial trajectory as a diffusion and derives the kill boundary; that is the gap this paper fills, and the learned rules are the ones its wavefront would predict. Yao et al. (2023) and Snell et al. (2024) frame the wider test-time-compute allocation problem; Malladi et al. (2026) train for coverage, shaping the distribution whose maximum is paid. The estimation side, predicting the expected maximum of a batch from few samples, is treated in a companion note (Cotton, 2026).

7 Numerical certificates

Every numerical claim is backed by a committed, seeded script in the repository[1]. In the committed example (horizon 1, initial intensity 100, budget 10, fallback 0, 100 time steps, 601 state points) the adaptive policy reaches an expected maximum of 1.9600, at a cost of exactly the budget and with a certificate below 2×10^{-6}, against 1.6429 for the best one-shot screening policy and 1.5086 for static random thinning. A 100,000-trial Monte Carlo puts the adaptive policy at 1.9587 with standard error 0.0021. The analytic gradient of (1) matches finite differences to 3×10^{-7}; spent path-time is monotone in the shadow price; refining the grid moves the objective by less than 0.08. A dependency-free JavaScript port runs the same solve in the browser and is held to the Python at stated tolerances by a committed parity test.

8 Closing

The mean-field, population-blind race is solved, certified, and cheap. Three problems remain open, in increasing order of difficulty: the value of population awareness beyond the one-step example of Section 2; the finite-n race under a pathwise hard budget, where the Poisson structure is lost; and the continuous-time limit, where we expect the wavefront cutoff to become a free boundary and the pivotal-value flux across it a collision local time. We leave to future work the correlated race, where paths share factors and the pruning decision must price a rival's strength.

References

Audibert, J.-Y., Bubeck, S., and Munos, R. (2010). Best arm identification in multi-armed bandits. In *COLT*.

Beneš, V. E., Shepp, L. A., and Witsenhausen, H. S. (1980). Some solvable stochastic control problems. *Stochastics*, 4(1):39–83.

Berestycki, J., Berestycki, N., and Schweinsberg, J. (2013). The genealogy of branching Brownian motion with absorption. *Annals of Probability*, 41(2):527–618.

[1] `github.com/microprediction/brownianbandit`; live demos at `brownianbandit.microprediction.org`.

Bi, J., Luo, T., Du, W., Tang, Z., and Wang, B. (2026). Cut your losses! Learning to prune paths early for efficient parallel reasoning. arXiv:2604.16029.

Brunet, E., and Derrida, B. (1997). Shift in the velocity of a front due to a cutoff. *Physical Review E*, 56(3):2597–2604.

Carmona, R., and Delarue, F. (2018). *Probabilistic Theory of Mean Field Games with Applications*, vols. I–II. Springer.

Carpentier, A., and Valko, M. (2014). Extreme bandits. In *NeurIPS*.

Chen, M., et al. (2021). Evaluating large language models trained on code. arXiv:2107.03374.

Cicirello, V. A., and Smith, S. F. (2005). The max k-armed bandit: a new model of exploration applied to search heuristic selection. In *AAAI*.

Cotton, P. (2026). Posterior-predictive pass-at-k. Working note, `brownianbandit.microprediction.org`.

Frank, M., and Wolfe, P. (1956). An algorithm for quadratic programming. *Naval Research Logistics Quarterly*, 3:95–110.

Gittins, J. C. (1979). Bandit processes and dynamic allocation indices. *Journal of the Royal Statistical Society, Series B*, 41(2):148–177.

Jaggi, M. (2013). Revisiting Frank–Wolfe: projection-free sparse convex optimization. In *ICML*.

Kalayci, Y., Raman, V., and Dughmi, S. (2025). Optimal stopping vs best-of-N for inference time optimization. arXiv:2510.01394.

Lasry, J.-M., and Lions, P.-L. (2007). Mean field games. *Japanese Journal of Mathematics*, 2(1):229–260.

Maillard, P. (2016). Speed and fluctuations of N-particle branching Brownian motion with spatial selection. *Probability Theory and Related Fields*, 166:1061–1173.

Malladi, S., Jelassi, S., Foster, D., Ash, J. T., and Krishnamurthy, A. (2026). TailSFT: filtered fine-tuning improves post-training performance. arXiv:2608.25756.

Nomand, P., Voss, E., Hale, M., and Reyes, S. (2026). Early verdicts, better budgets: sequential adaptive rollout allocation for compute-efficient RLVR. arXiv:2607.26253.

Snell, C., Lee, J., Xu, K., and Kumar, A. (2024). Scaling LLM test-time compute optimally can be more effective than scaling model parameters. arXiv:2408.03314.

Sun, H., Haider, M., Zhang, R., Yang, H., Qiu, J., Yin, M., Wang, M., Bartlett, P., and Zanette, A. (2024). Fast best-of-N decoding via speculative rejection. In *NeurIPS*.

Weitzman, M. L. (1979). Optimal search for the best alternative. *Econometrica*, 47(3):641–654.

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., and Narasimhan, K. (2023). Tree of Thoughts: deliberate problem solving with large language models. In *NeurIPS*.